```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (HH)
  Ansatz          : warm: None; ADAPT: legacy; final: matched-family replay
  Drive enabled   : False

  t (hopping)     : 1.0
  U (interaction) : 2.0
  dv (disorder)   : 0.0

  L_values        : 2,3
  omega0          : 1.0
  g_ep            : 1.0
  n_ph_max        : 1
  ordering        : blocked
  boundary        : periodic
  dynamics_pages  : suzuki2,cfqm4 macro-step only

  Command:
    /opt/anaconda3/bin/python pipelines/hardcoded/hh_staged_circuit_report.py --smoke-test-intentionally-weak
    --warm-reps 1 --warm-restarts 1 --warm-maxiter 1 --adapt-continuation-mode legacy --adapt-pool paop_lf_std
    --adapt-disable-hh-seed --adapt-max-depth 2 --adapt-maxiter 1 --adapt-eps-grad 1000000 --adapt-eps-energy
    1000000 --final-reps 1 --final-restarts 1 --final-maxiter 1 --replay-continuation-mode legacy --trotter-
    steps 1 --num-times 2 --output-pdf artifacts/pdf/hh_staged_circuit_report_L2_L3_smoke.pdf --tag
    hh_staged_circuit_smoke_legacy_nohhseed

============================================================
```

```
Combined HH staged circuit report

L values: 2, 3
Each L section includes:
- parameter manifest
- stage summary
- warm HH-HVA circuit
- ADAPT circuit
- matched-family replay circuit
- Suzuki2 macro-step circuit
- CFQM4 macro-step circuit

Representative pages preserve PauliEvolutionGate blocks.
Expanded pages decompose one level only for readability.
```

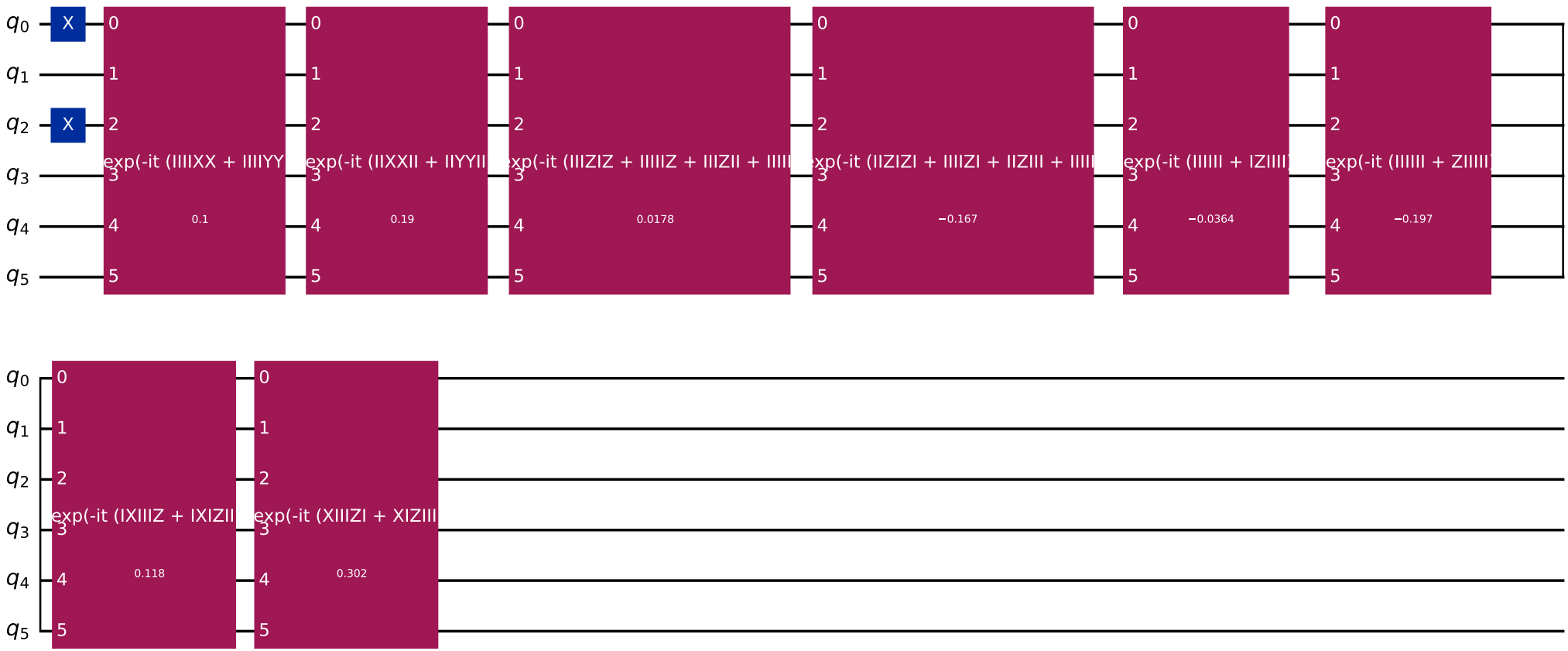
```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (HH)
  Ansatz          : warm: hh_hva_ptw; ADAPT: legacy; final: matched-family replay
  Drive enabled   : False

  t (hopping)     : 1.0
  U (interaction) : 2.0
  dv (disorder)   : 0.0

  L               : 2
  omega0          : 1.0
  g_ep            : 1.0
  n_ph_max        : 1
  boundary        : periodic
  ordering        : blocked
  warm_reps       : 1
  adapt_max_depth : 2
  replay_reps     : 1
  t_final         : 10.0
  trotter_steps   : 1
  num_times       : 2

  Command:
    /opt/anaconda3/bin/python pipelines/hardcoded/hh_staged_circuit_report.py --smoke-test-intentionally-weak
    --warm-reps 1 --warm-restarts 1 --warm-maxiter 1 --adapt-continuation-mode legacy --adapt-pool paop_lf_std
    --adapt-disable-hh-seed --adapt-max-depth 2 --adapt-maxiter 1 --adapt-eps-grad 1000000 --adapt-eps-energy
    1000000 --final-reps 1 --final-restarts 1 --final-maxiter 1 --replay-continuation-mode legacy --trotter-
    steps 1 --num-times 2 --output-pdf artifacts/pdf/hh_staged_circuit_report_L2_L3_smoke.pdf --tag
    hh_staged_circuit_smoke_legacy_nohhseed

============================================================
```

```
HH staged circuit report, L=2

Warm: E=3.03024280324 exact=-0.389553103297 |dE|=3.419796e+00
ADAPT: depth=2 pool=paop_lf_std stop=max_depth |dE|=2.331930e+00
Replay: E=1.94237660187 exact=-0.389553103297 |dE|=2.331930e+00

Section semantics
- Representative view keeps high-level PauliEvolutionGate blocks.
- Expanded view performs one decomposition pass to expose term-level layers.
- Dynamics pages show one macro-step only; proxy totals summarize the full repeat count.
```

```
L=2 warm HH-HVA

Circuit summary
  - num_qubits: 6
  - depth: 9
  - size: 10
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 8, "x": 2}

Metadata
  - ansatz: hh_hva_ptw
  - reps: 1
  - energy: 3.03024280324
  - exact_energy: -0.389553103297
  - delta_abs: 3.41979590654

Notes
  - Representative view keeps PauliEvolutionGate blocks intact.
  - Expanded view applies one circuit-definition decomposition pass.
```

L=2 warm HH-HVA

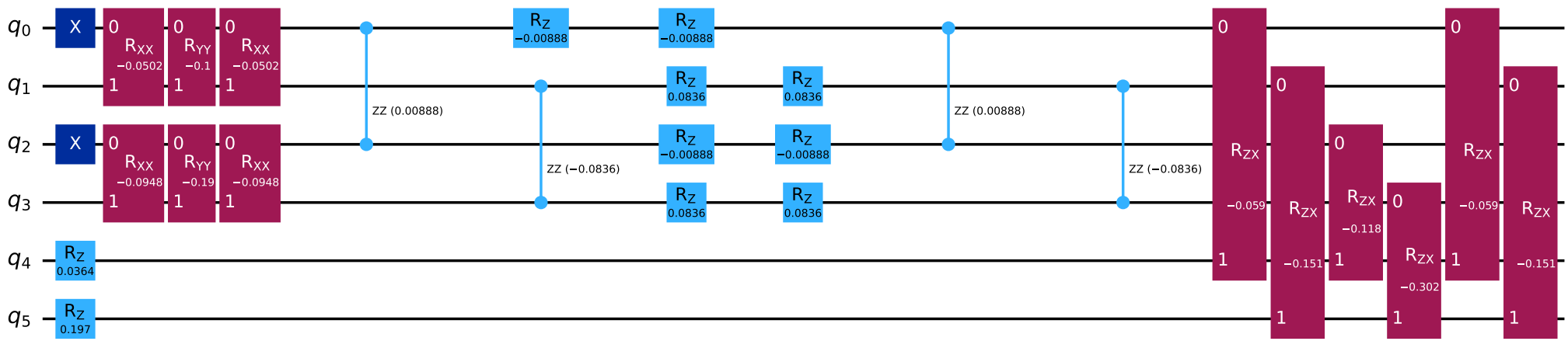
Representative view




```
Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.
```

L=2 warm HH-HVA

Expanded view

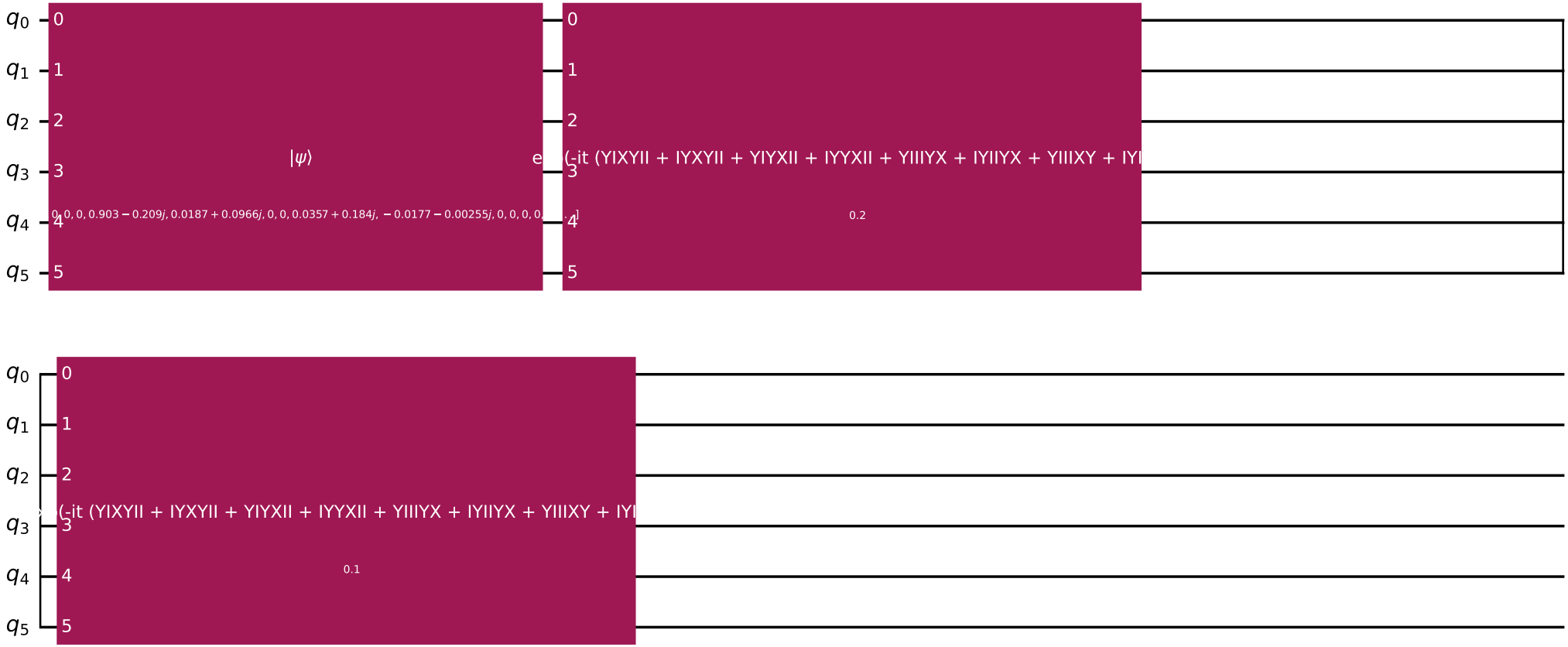


Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.

```
L=2 ADAPT-VQE

Circuit summary
  - num_qubits: 6
  - depth: 3
  - size: 3
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 2, "initialize": 1}

Metadata
  - depth: 2
  - pool_type: paop_lf_std
  - continuation_mode: legacy
  - energy: 1.94237660187
  - exact_energy: -0.389553103297
  - stop_reason: max_depth

Notes
  - Circuit uses the actual selected ADAPT generators and optimized theta values.
  - Reference state is the warm-stage output handed to ADAPT.
```

L=2 ADAPT-VQE

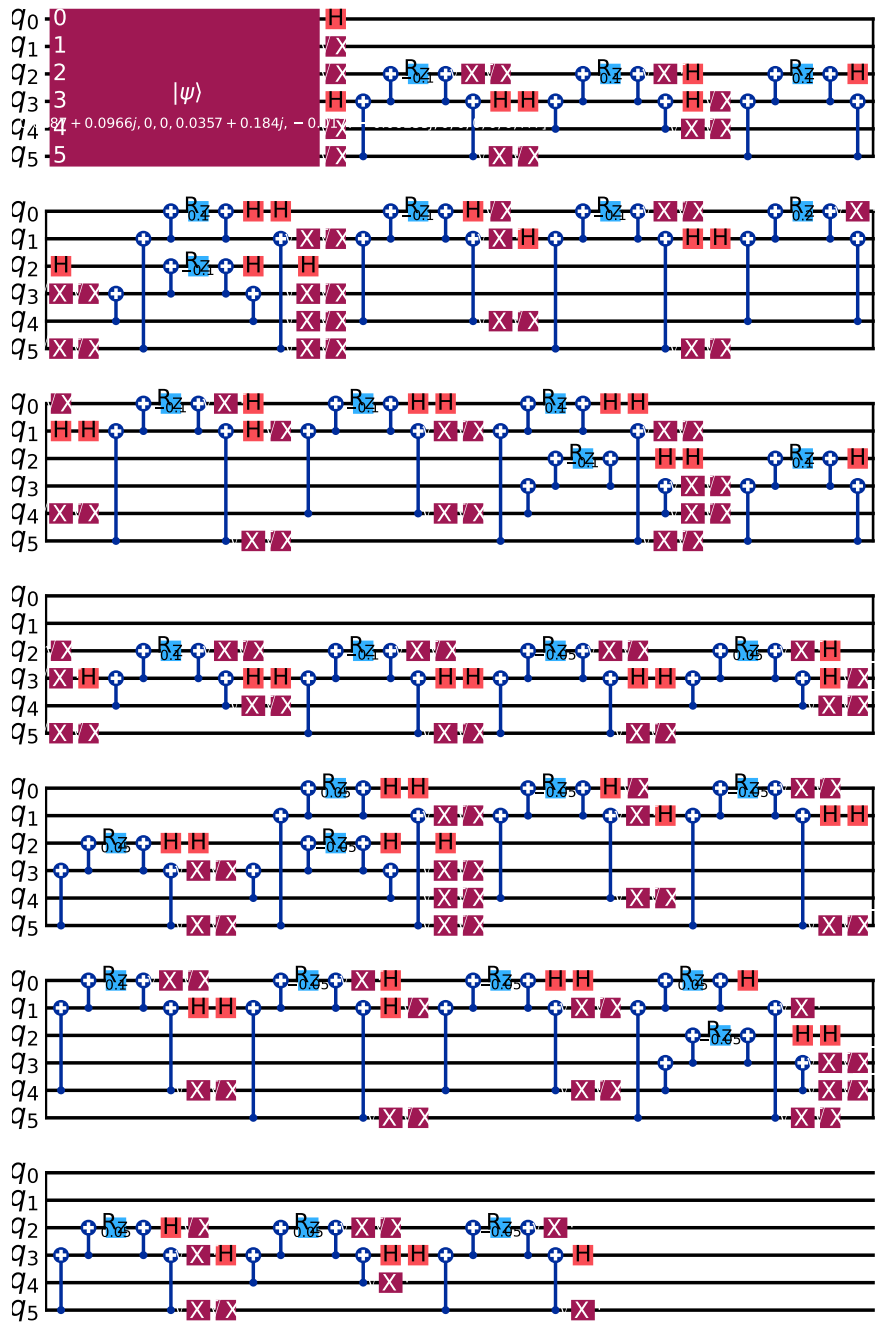
Representative view



Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

L=2 ADAPT-VQE

Expanded view



Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

```
L=2 matched-family replay

Circuit summary
  - num_qubits: 6
  - depth: 3
  - size: 3
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 2, "initialize": 1}

Metadata
  - family_info: {"fallback_family": "full_meta", "fallback_used": false, "requested": "match_adapt",
    "resolution_source": "adapt_vqe.pool_type", "resolved": "paop_lf_std", "warning": null}
  - reps: 1
  - seed_policy: residual_only
  - energy: 1.94237660187
  - exact_energy: -0.389553103297
  - stop_reason: converged

Notes
  - Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
  - This is the conventional non-ADAPT VQE stage in the staged HH workflow.
```

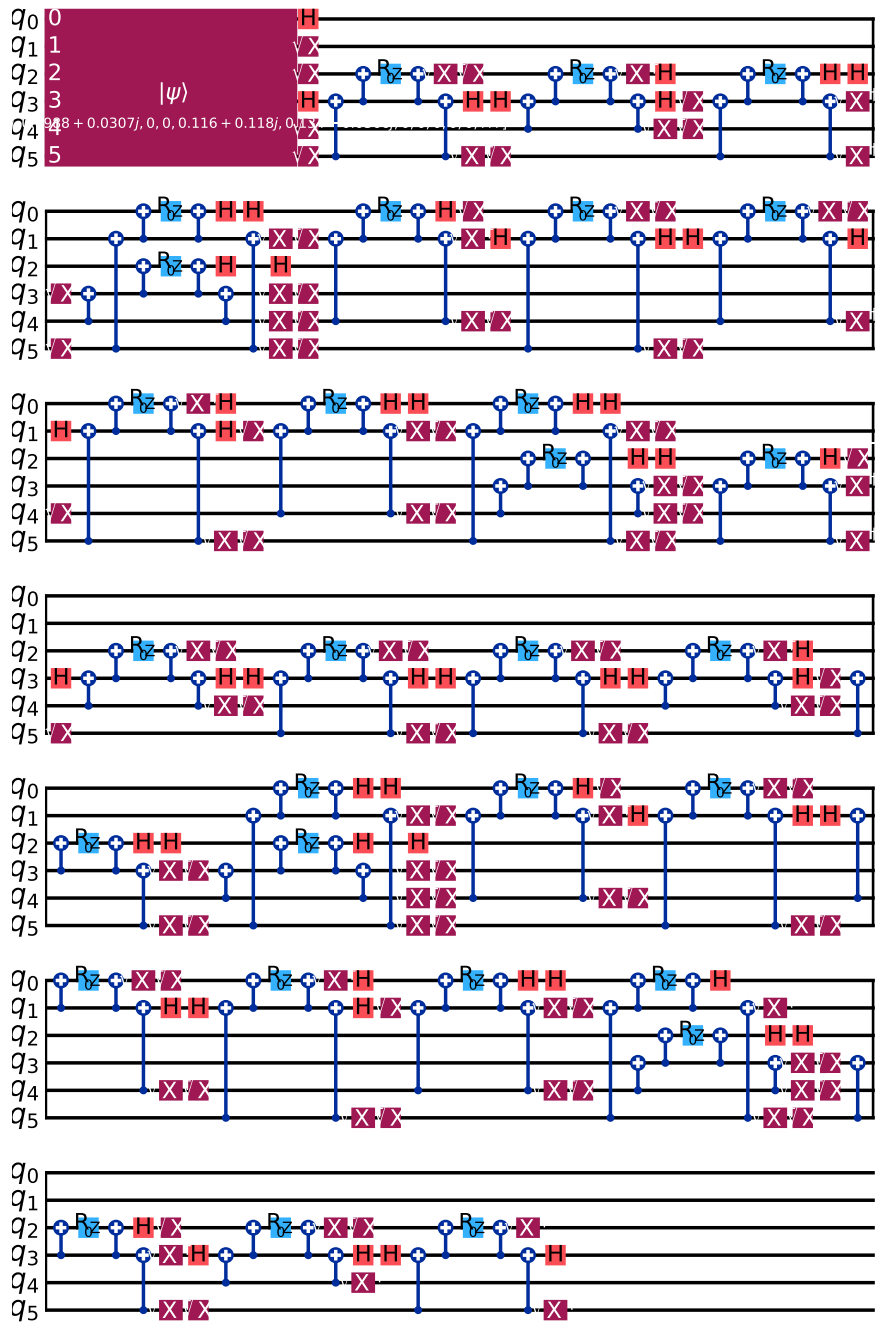
L=2 matched-family replay

Representative view

Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

L=2 matched-family replay

Expanded view



Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

```
L=2 SUZUKI2 dynamics macro-step

Circuit summary
  - num_qubits: 6
  - depth: 4
  - size: 4
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 3, "initialize": 1}

Metadata
  - repeat_count: 1
  - macro_step_time: 10
  - t_final: 10
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 40, "sq_proxy_total": 82, "term_exp_count_total": 34}
  - proxy_total: {"cx_proxy_total": 40, "sq_proxy_total": 82, "term_exp_count_total": 34}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

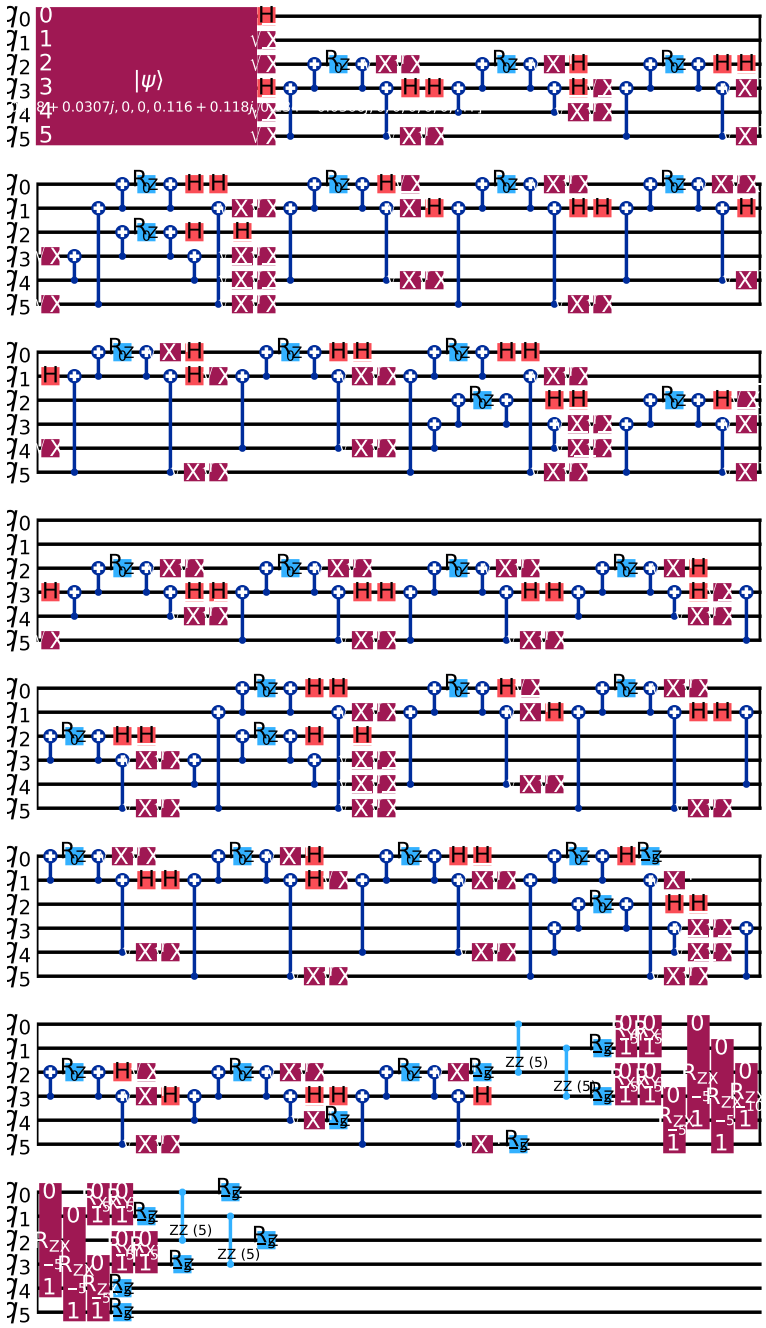
L=2 SUZUKI2 dynamics macro-step

Representative macro-step

```
Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=2 SUZUKI2 dynamics macro-step

Expanded macro-step

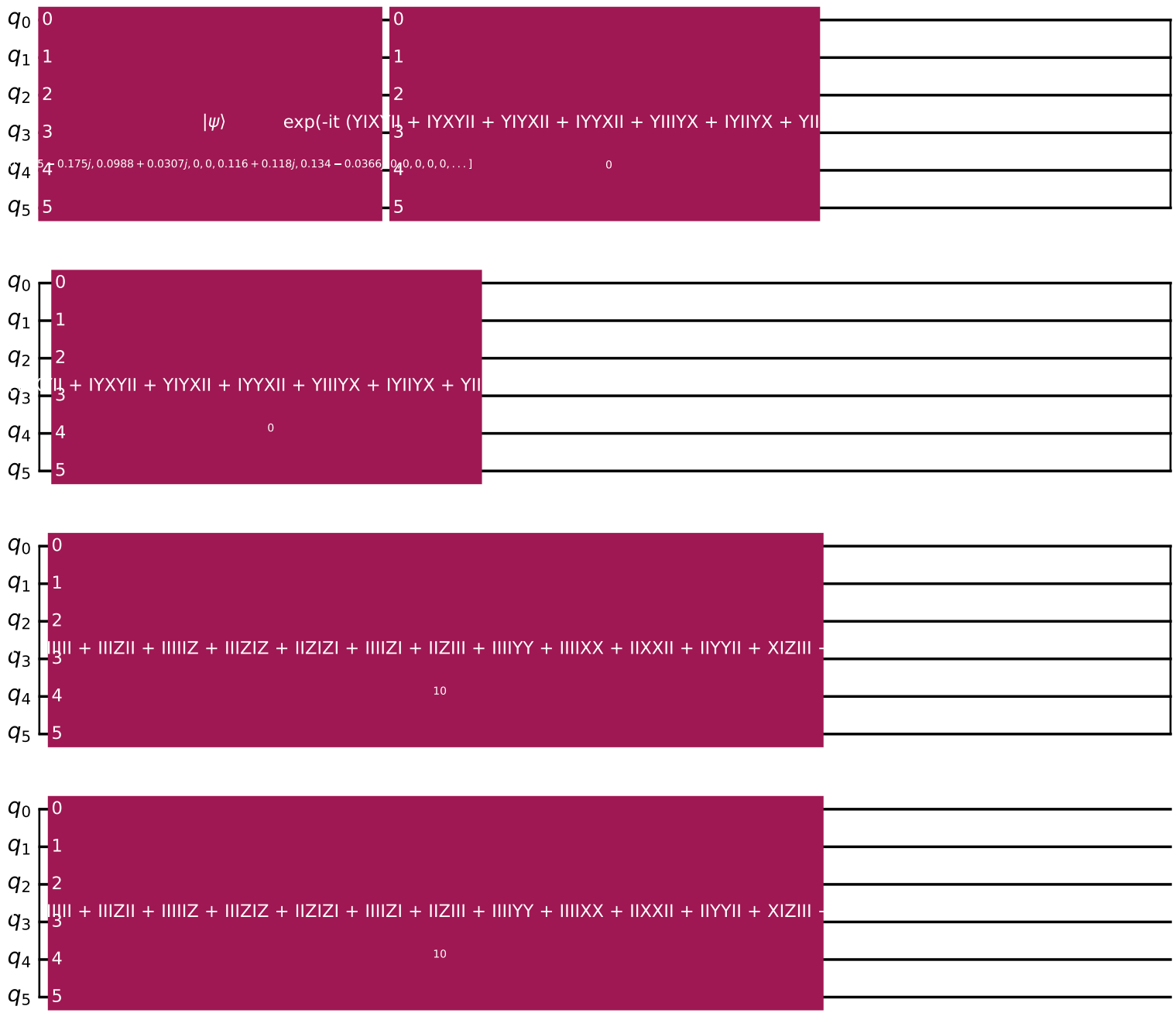


Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

```
L=2 CFQM4 dynamics macro-step

Circuit summary
  - num_qubits: 6
  - depth: 5
  - size: 5
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 4, "initialize": 1}

Metadata
  - repeat_count: 1
  - macro_step_time: 10
  - t_final: 10
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 80, "sq_proxy_total": 164, "term_exp_count_total": 68}
  - proxy_total: {"cx_proxy_total": 80, "sq_proxy_total": 164, "term_exp_count_total": 68}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=2 CFQM4 dynamics macro-step

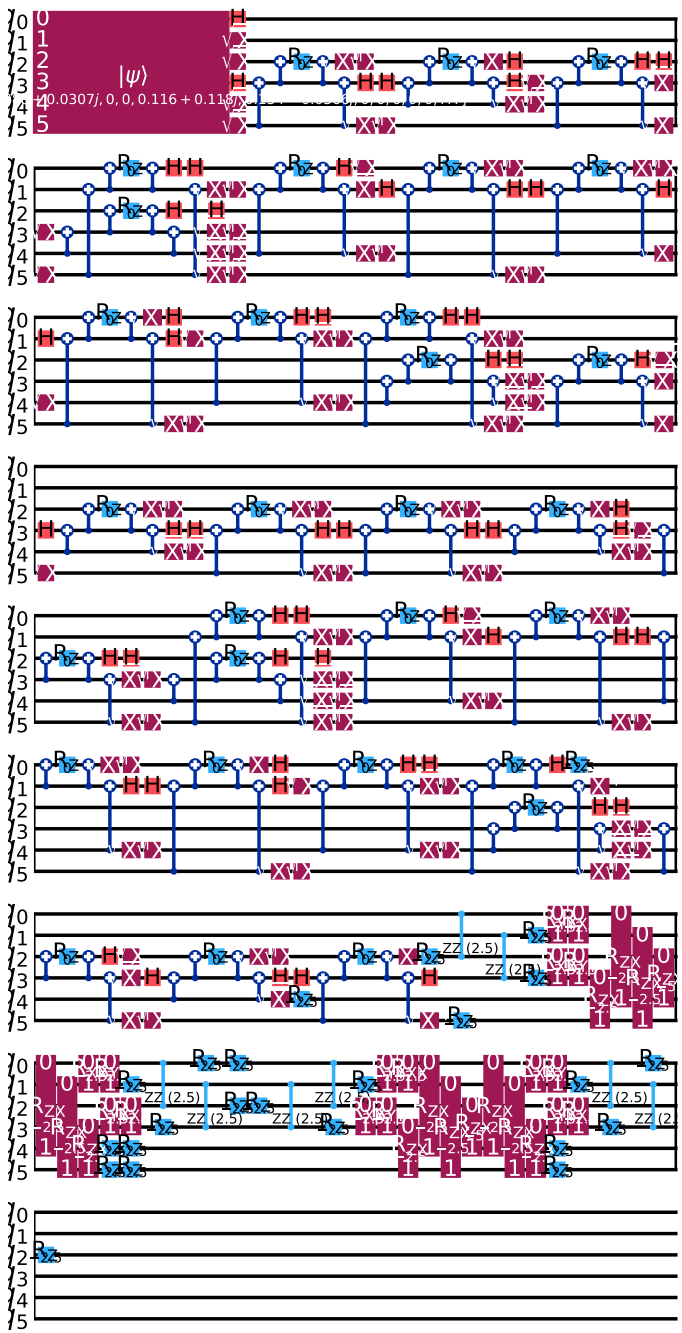
Representative macro-step




```
Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=2 CFQM4 dynamics macro-step

Expanded macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.

```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (HH)
  Ansatz          : warm: hh_hva_ptw; ADAPT: legacy; final: matched-family replay
  Drive enabled   : False

  t (hopping)     : 1.0
  U (interaction) : 2.0
  dv (disorder)   : 0.0

  L               : 3
  omega0          : 1.0
  g_ep            : 1.0
  n_ph_max        : 1
  boundary        : periodic
  ordering        : blocked
  warm_reps       : 1
  adapt_max_depth : 2
  replay_reps     : 1
  t_final         : 15.0
  trotter_steps   : 1
  num_times       : 2

  Command:
    /opt/anaconda3/bin/python pipelines/hardcoded/hh_staged_circuit_report.py --smoke-test-intentionally-weak
    --warm-reps 1 --warm-restarts 1 --warm-maxiter 1 --adapt-continuation-mode legacy --adapt-pool paop_lf_std
    --adapt-disable-hh-seed --adapt-max-depth 2 --adapt-maxiter 1 --adapt-eps-grad 1000000 --adapt-eps-energy
    1000000 --final-reps 1 --final-restarts 1 --final-maxiter 1 --replay-continuation-mode legacy --trotter-
    steps 1 --num-times 2 --output-pdf artifacts/pdf/hh_staged_circuit_report_L2_L3_smoke.pdf --tag
    hh_staged_circuit_smoke_legacy_nohhseed

============================================================
```

```
HH staged circuit report, L=3

Warm: E=3.32234716999 exact=-0.6988083851 |dE|=4.021156e+00
ADAPT: depth=2 pool=paop_lf_std stop=max_depth |dE|=3.387631e+00
Replay: E=2.68882245725 exact=-0.6988083851 |dE|=3.387631e+00

Section semantics
- Representative view keeps high-level PauliEvolutionGate blocks.
- Expanded view performs one decomposition pass to expose term-level layers.
- Dynamics pages show one macro-step only; proxy totals summarize the full repeat count.
```

```
L=3 warm HH-HVA

Circuit summary
  - num_qubits: 9
  - depth: 16
  - size: 18
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 15, "x": 3}

Metadata
  - ansatz: hh_hva_ptw
  - reps: 1
  - energy: 3.32234716999
  - exact_energy: -0.6988083851
  - delta_abs: 4.02115555509

Notes
  - Representative view keeps PauliEvolutionGate blocks intact.
  - Expanded view applies one circuit-definition decomposition pass.
```

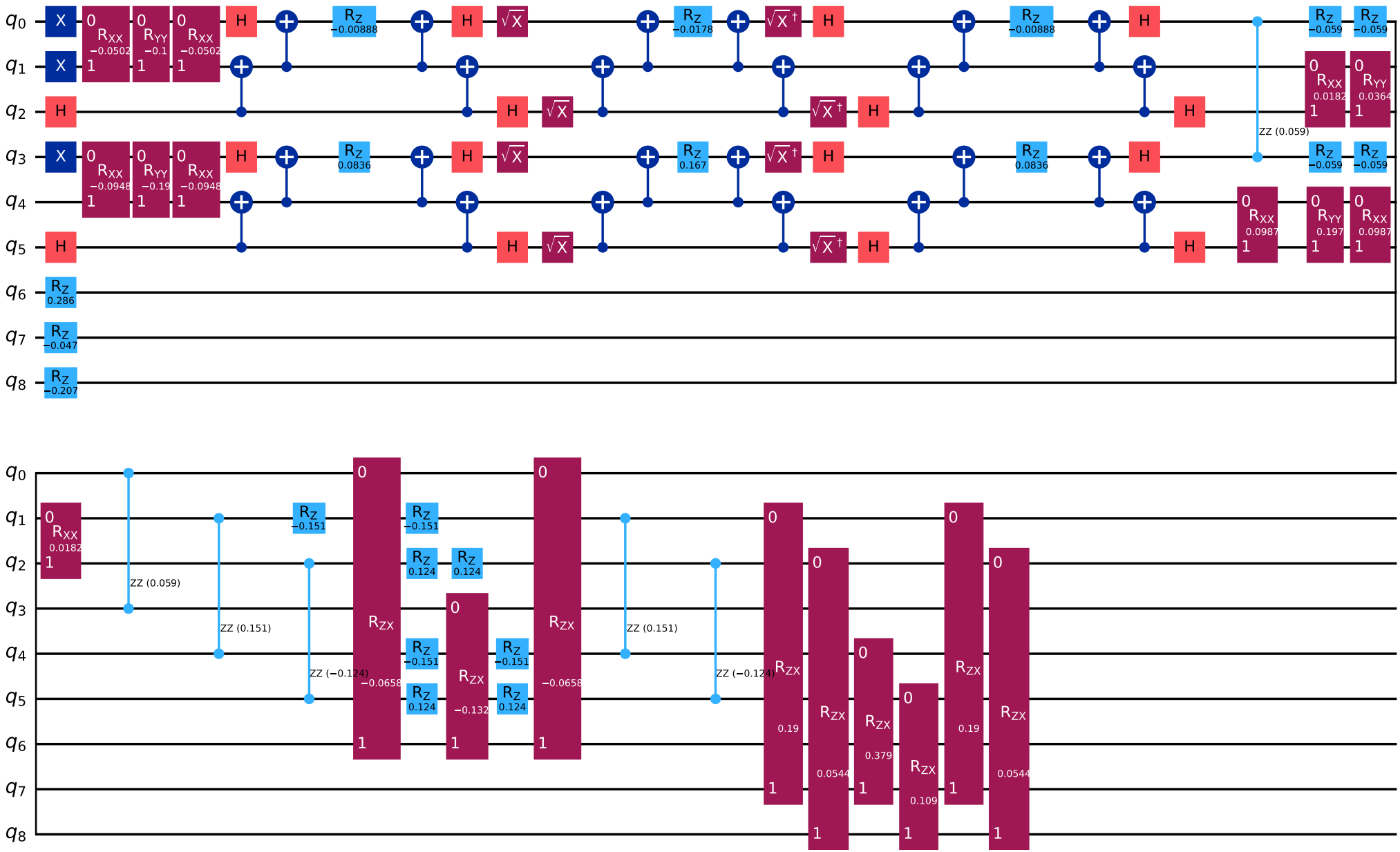
L=3 warm HH-HVA

Representative view

Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.

L=3 warm HH-HVA

Expanded view



Representative view keeps PauliEvolutionGate blocks intact.
Expanded view applies one circuit-definition decomposition pass.

```
L=3 ADAPT-VQE

Circuit summary
  - num_qubits: 9
  - depth: 3
  - size: 3
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 2, "initialize": 1}

Metadata
  - depth: 2
  - pool_type: paop_lf_std
  - continuation_mode: legacy
  - energy: 2.68882245725
  - exact_energy: -0.6988083851
  - stop_reason: max_depth

Notes
  - Circuit uses the actual selected ADAPT generators and optimized theta values.
  - Reference state is the warm-stage output handed to ADAPT.
```

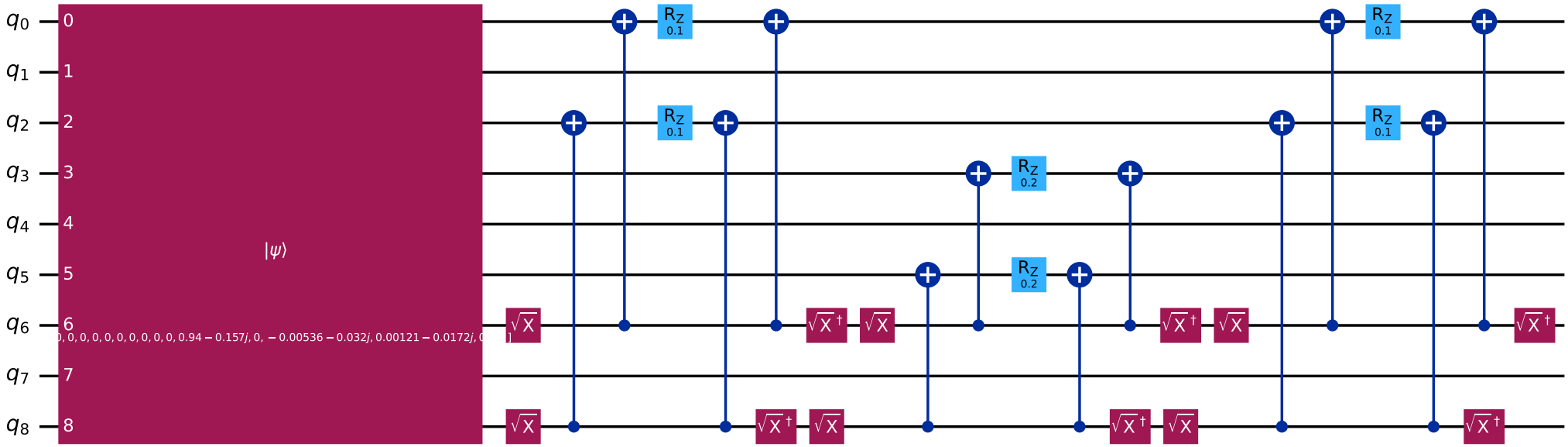
L=3 ADAPT-VQE

Representative view

Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

L=3 ADAPT-VQE

Expanded view



Circuit uses the actual selected ADAPT generators and optimized theta values.
Reference state is the warm-stage output handed to ADAPT.

```
L=3 matched-family replay

Circuit summary
  - num_qubits: 9
  - depth: 3
  - size: 3
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 2, "initialize": 1}

Metadata
  - family_info: {"fallback_family": "full_meta", "fallback_used": false, "requested": "match_adapt",
    "resolution_source": "adapt_vqe.pool_type", "resolved": "paop_lf_std", "warning": null}
  - reps: 1
  - seed_policy: residual_only
  - energy: 2.68882245725
  - exact_energy: -0.6988083851
  - stop_reason: converged

Notes
  - Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
  - This is the conventional non-ADAPT VQE stage in the staged HH workflow.
```

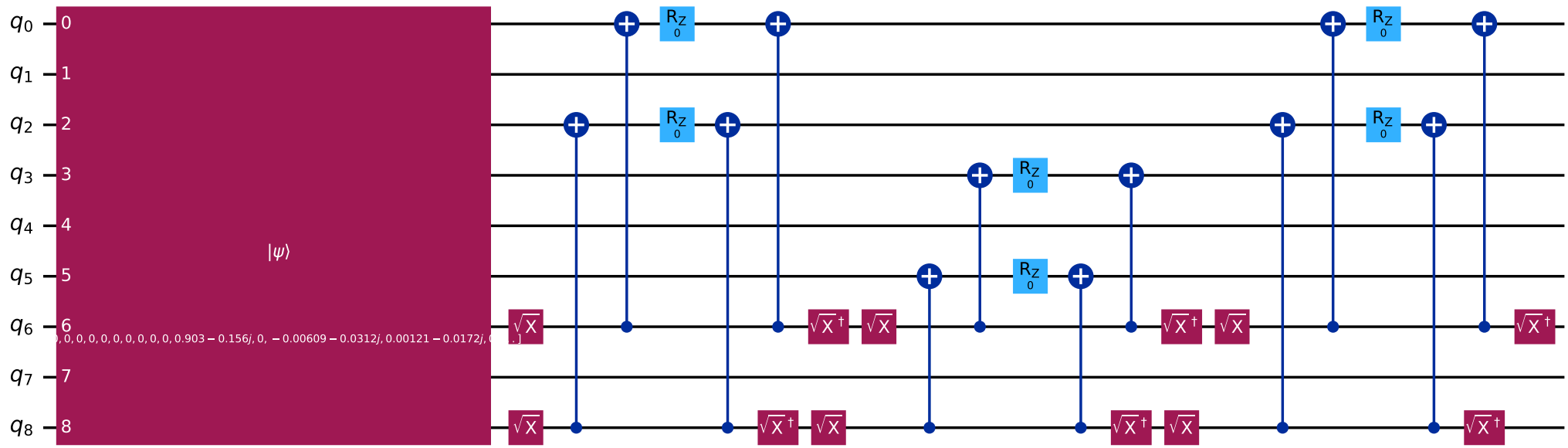
L=3 matched-family replay

Representative view

Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

L=3 matched-family replay

Expanded view

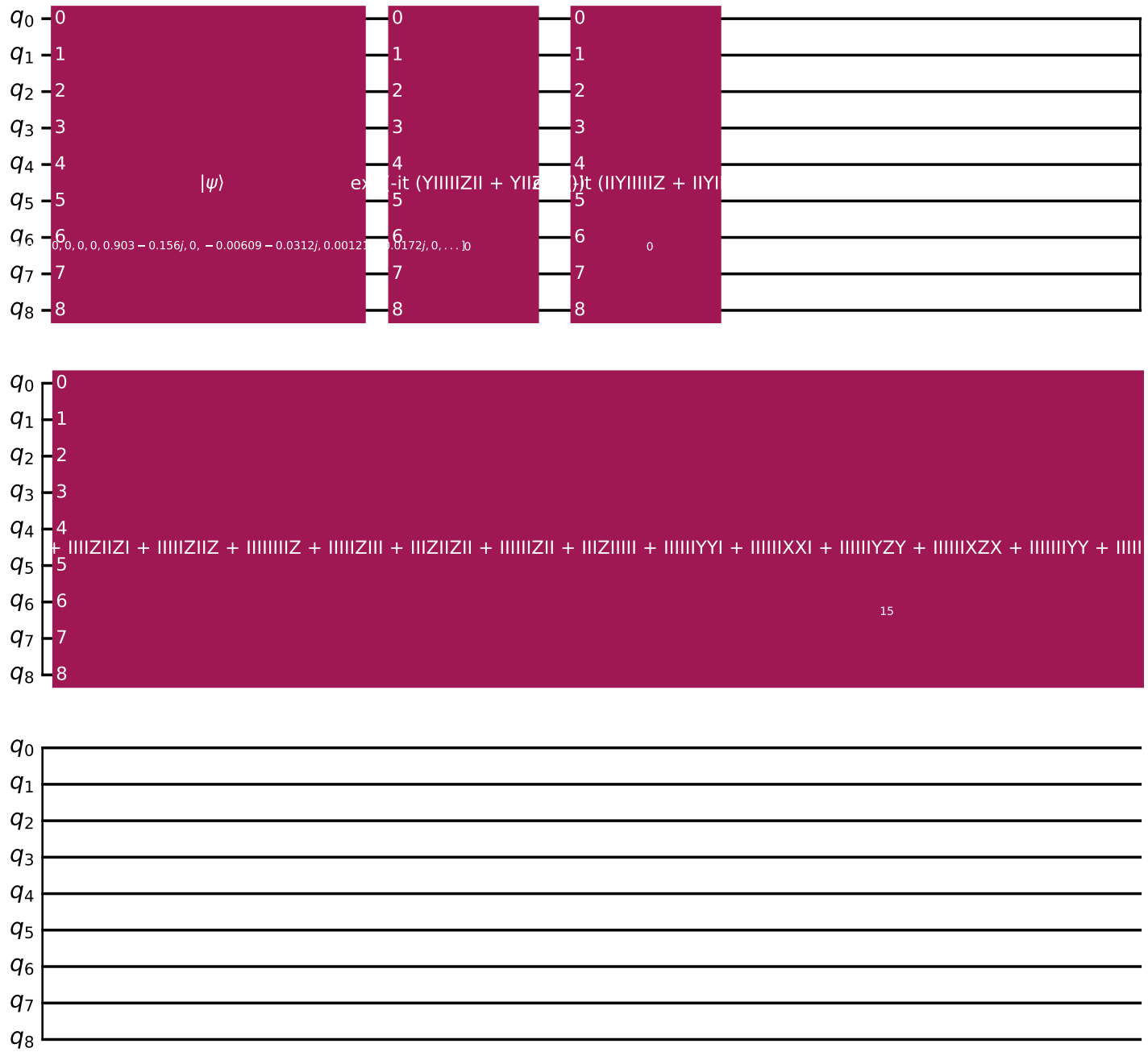


Replay circuit uses the matched ADAPT-family ansatz with the final optimized replay theta values.
This is the conventional non-ADAPT VQE stage in the staged HH workflow.

```
L=3 SUZUKI2 dynamics macro-step

Circuit summary
  - num_qubits: 9
  - depth: 4
  - size: 4
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 3, "initialize": 1}

Metadata
  - repeat_count: 1
  - macro_step_time: 15
  - t_final: 15
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 100, "sq_proxy_total": 182, "term_exp_count_total": 62}
  - proxy_total: {"cx_proxy_total": 100, "sq_proxy_total": 182, "term_exp_count_total": 62}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=3 SUZUKI2 dynamics macro-step

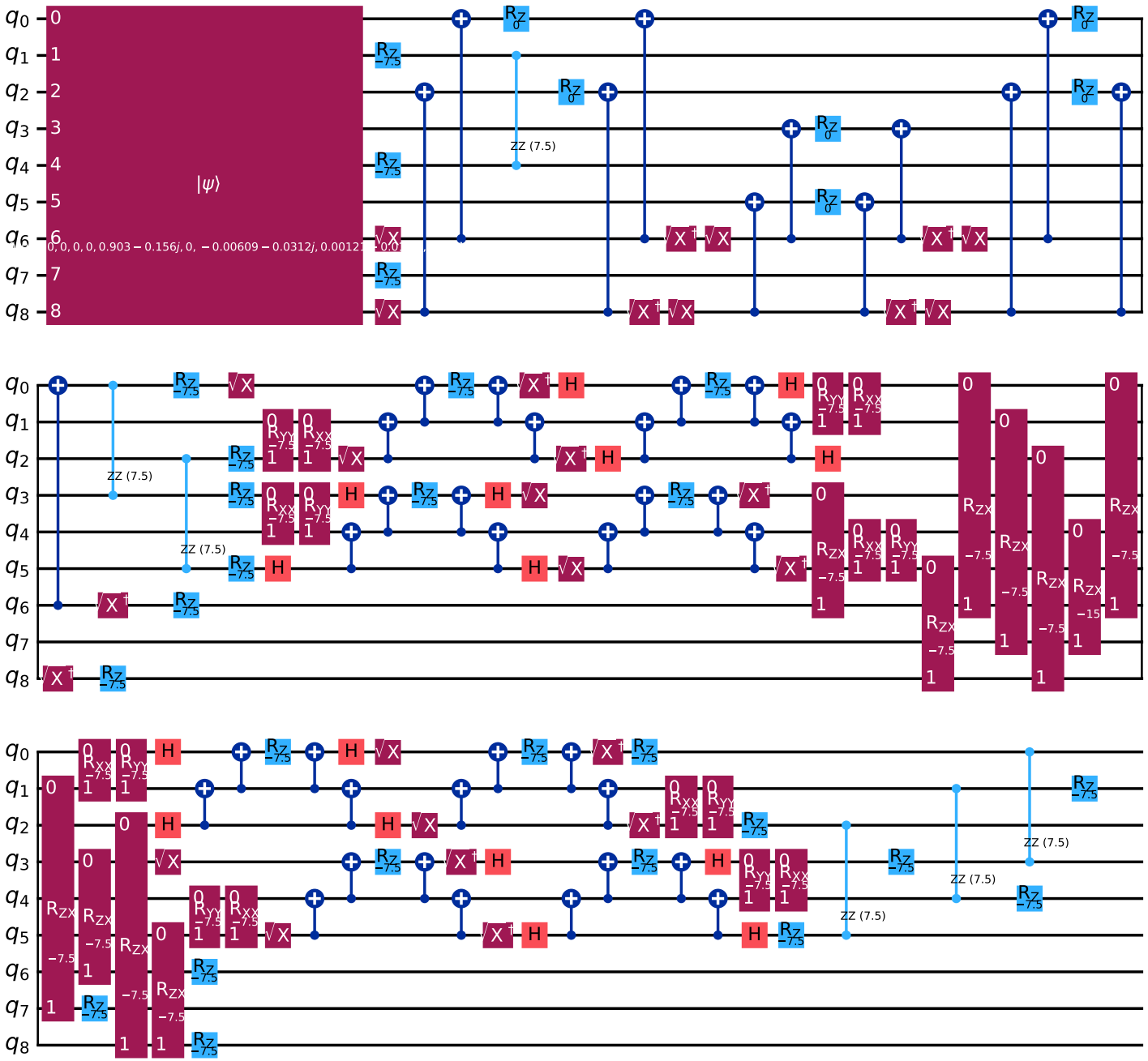
Representative macro-step




```
Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=3 SUZUKI2 dynamics macro-step

Expanded macro-step

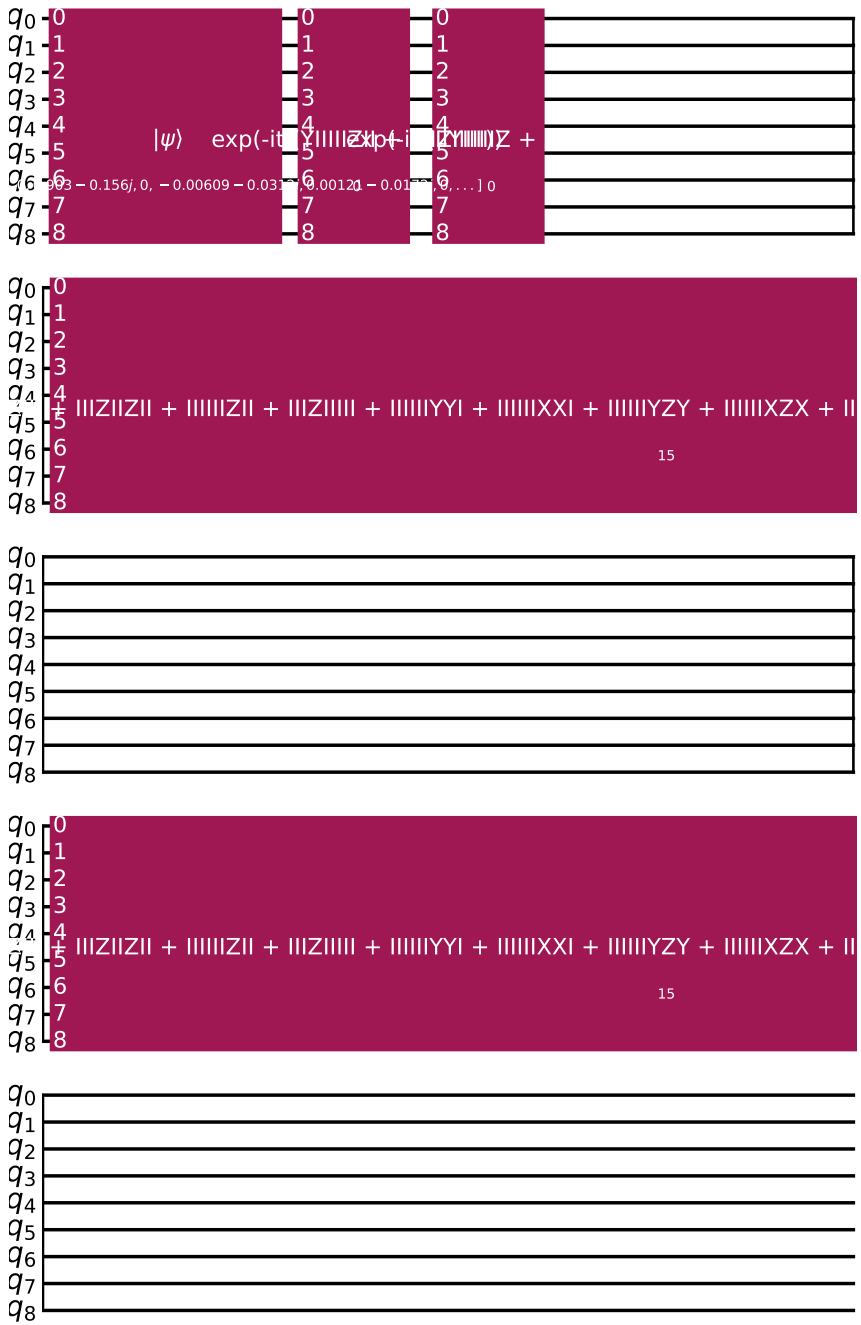



```
Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

```
L=3 CFQM4 dynamics macro-step

Circuit summary
  - num_qubits: 9
  - depth: 5
  - size: 5
  - num_parameters: 0
  - count_ops: {"PauliEvolution": 4, "initialize": 1}

Metadata
  - repeat_count: 1
  - macro_step_time: 15
  - t_final: 15
  - drive_enabled: False
  - drive_profile: n/a
  - proxy_macro: {"cx_proxy_total": 200, "sq_proxy_total": 364, "term_exp_count_total": 124}
  - proxy_total: {"cx_proxy_total": 200, "sq_proxy_total": 364, "term_exp_count_total": 124}

Notes
  - Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
  - Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=3 CFQM4 dynamics macro-step

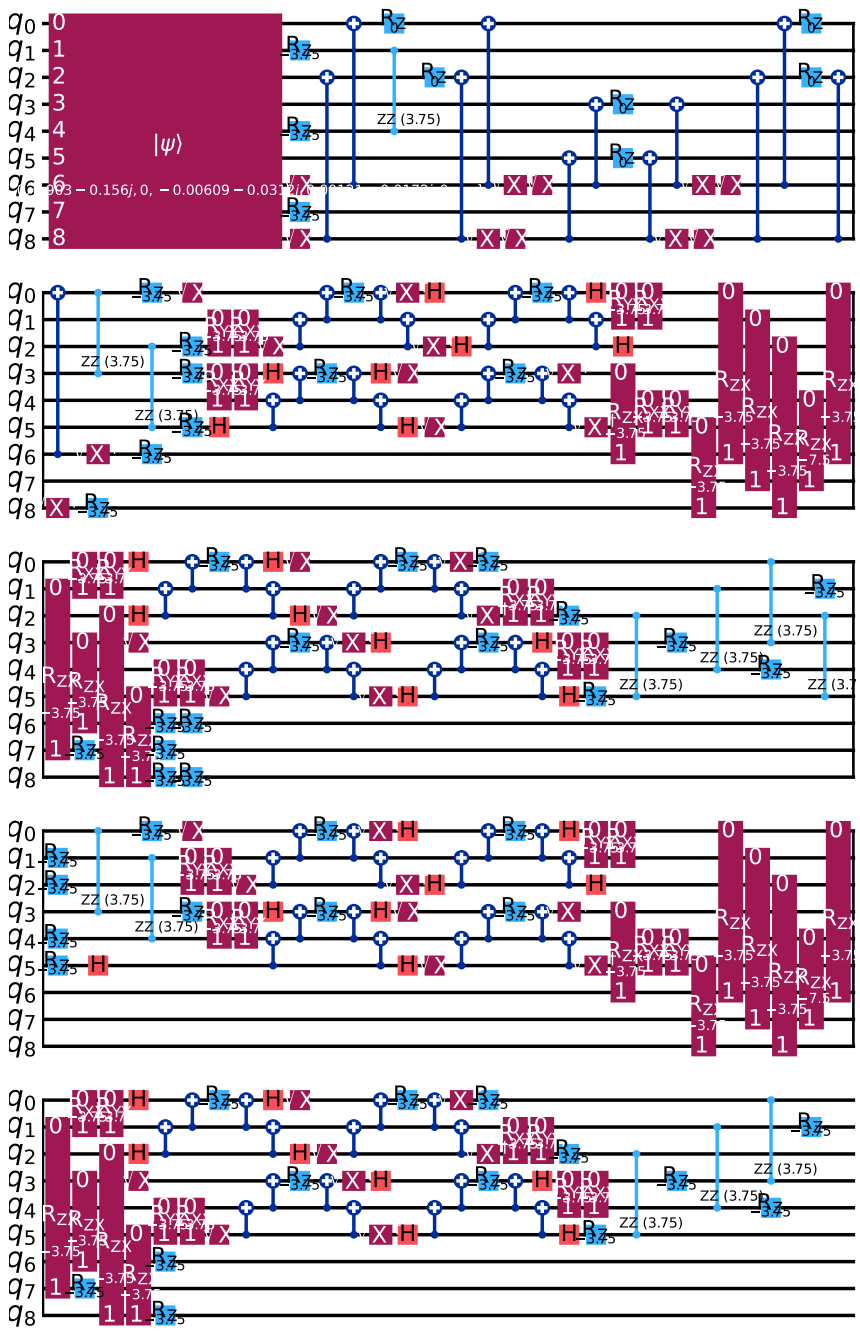
Representative macro-step




```
Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.
```

L=3 CFQM4 dynamics macro-step

Expanded macro-step



Circuit shows one representative macro-step only; repeat_count gives the full unrolled count.
Expanded view decomposes only the PauliEvolutionGate layers, not the full repeated trajectory.